Lesson for the course "Human-Computer Interaction"

Hands-on workshop on the design phase: from Personas to low-fidelity prototypes

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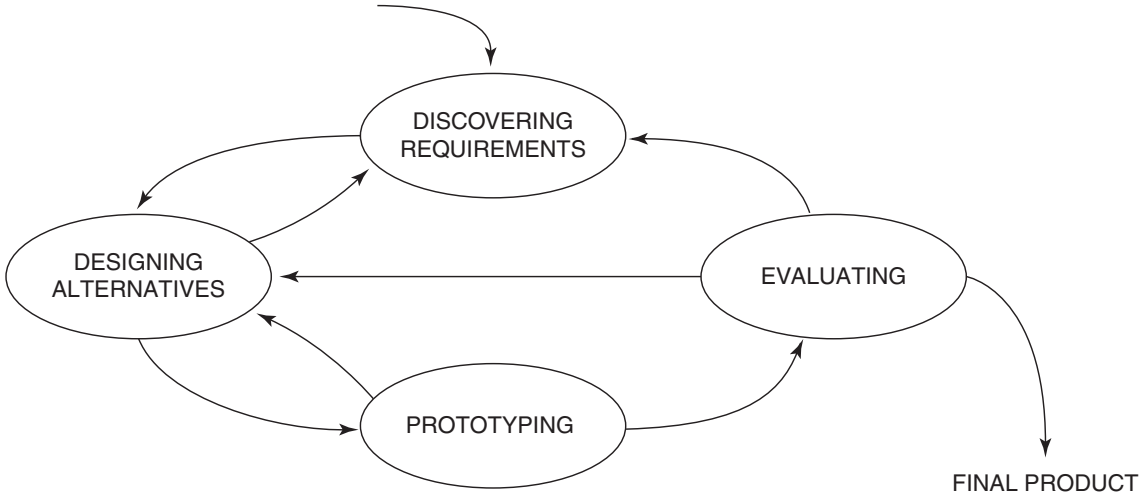
Workshop topics

- Personas
- Brainstorming
- Concept definition
- Envisioning scenarios
- Low-fidelity prototypes

- You will be working on the design problem of **your project**
- Please notice: we will not complete all the work that needs to be done for each phase; the purpose of this lesson is to see how to structure properly the work you need to do for your project

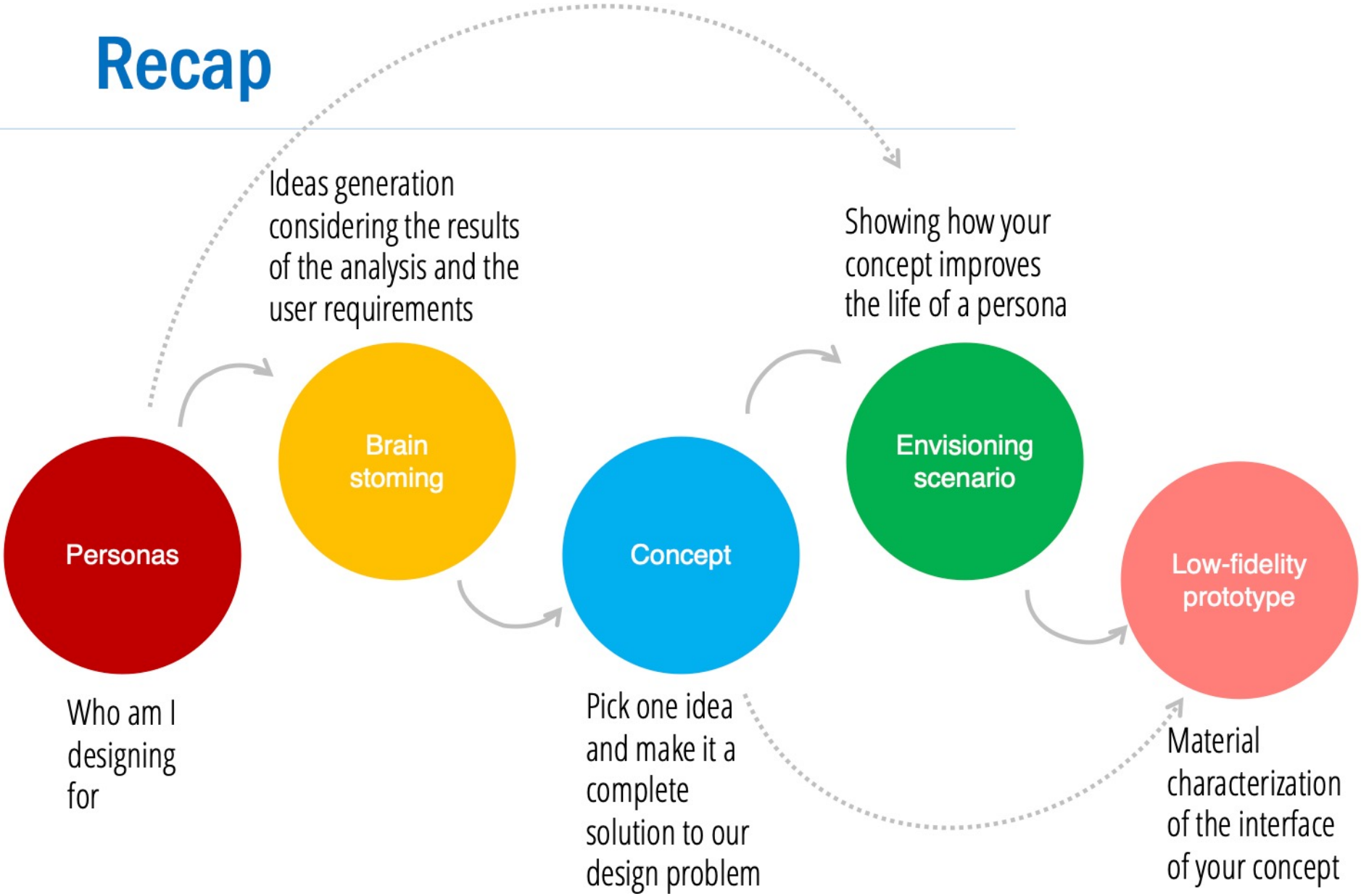
Recap

- We are in the **design phase** of the Interaction Design process



- We arrive at the design phase after we have
  - Identified a design problem
  - Conducted desk research and PACT analysis
  - Collected data with the stakeholders involved in the design problem
  - Understood what are their problems, needs, desires
- In the design phase we start thinking of what solutions we could provide to our stakeholders to improve their lives

Recap


Ideas generation considering the results of the analysis and the user requirements
Showing how your concept improves the life of a persona
Personas
Brain stoming
Concept
Envisioning scenario
Low-fidelity prototype
Who am I designing for
Pick one idea and make it a complete solution to our design problem
Material characterization of the interface of your concept

Personas

- Personas are fictional characters
- **Each persona represents a category of potential users**
- We create them to summarize the goals, the attitude, the needs and skills of the different categories of people we are designing for

Example of Persona

Clark Andrews

AGE	26
OCCUPATION	Software Developer
STATUS	Single
LOCATION	San Jose, CA
TIER	Experiment Hacker
ARCHETYPE	The Computer Nerd

Friendly | Clever | Go-Getter



"I feel like there's a smarter way for me to transition into a healthier lifestyle."

Motivations

- Incentive
- Fear
- Achievement
- Growth
- Power
- Social

Goals

- To cut down on unhealthy eating and drinking habits
- To measure multiple aspects of life more scientifically
- To set goals and see and make positive impacts on his life

Frustrations

- Unfamiliar with wearable technology
- Saturated tracking market
- Manual tracking is too time consuming

Bio

Aaron is a systems software developer, a "data junkie" and for the past couple years, has been very interested in tracking aspects of his health and performance. Aaron wants to track his mood, happiness, sleep quality and how his eating and exercise habits affects his well being. Although he only drinks occasionally with friends on the weekend, he would like to cut down on alcohol intake.

Personality

- Extrovert / Introvert
- Sensing / Intuition
- Thinking / Feeling
- Judging / Perceiving

Technology

- IT & Internet
- Software
- Mobile Apps
- Social Networks

Brands

Example of Persona



Nerdy Nina

"The book is way better than the movie!"

#booklover
#bookaddict
#booknerdproblems

DEMOGRAPHICS

Age:	25
Location:	Sao Paulo, Brazil
Education:	Software Engineer
Job:	Q/A at Indie Game Company
Family:	Lives with her boyfriend

TECH

Internet	●●●●●
Social Networks	●●○○○
Messaging	●●●○○
Games	●●●●●
Online Shopping	●●●●○

GOALS

- Discovering new books / authors to read
- Finding unique stories
- Cataloging book collection

FRUSTRATIONS

- Keeping track of different series
- Forgetting a book launch date
- Finding space for more books

READING HABITS

- Fast pace reader
- Never lends books
- Likes hardcovers and boxed collections
- Pre-order books to get them first
- Reads eBooks, but prefer physical copies
- Always finishes a book
- Loves binge reading and re-reading

FAVORITE BOOKS



American Gods
Neil Gaiman



Harry Potter
J.K. Rowling



Ready Player One

Example of Persona



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Demographics

- Age: 32
- Location: Chongqing, China
- Job: Purchasing department manager

Goals

- Accumulate a large number of professional vocabulary
- Read bussiness letter with no hint or barrier

Learning habits

- Using fragmentation time to learn English
- Reads eBooks but prefer the paper book

Frustrations

- Can't recognize some professional words instantly
- Always fogot some words learned before

Phone usage

Social networks

Messaging

Learning

Example of Persona



Rahul Bajaj
Mumbai, India

Gender : **Male**

Age: **34**

Status: **Single**

Education: **MBA**

Profession: **Job**

Archetype: **Explorer**

Bio

Rahul Bajaj is a tech geek who loves reading and giving his opinion on the new things happening around. He lives in Mumbai, India. He has a lot of friends and he visits them usually on weekends. He is a big fan of discovery TV channel. Whenever he has time, he watched various shows on Discovery such as Man Hunt.

He uses various apps such as the verge and NDTV to get updates from all over the glob reading new discovery in space science and techologies. However, he feels frustred specially when he needs to switch apps to read news from different domain. For example, he can get updates regarding technoogy innovations on the verge whereas he uses NDTV for speifially electronic gadgets.

He is alway hunting for a single app that can provide up to date news of new discovery.

Personality

Social — Ego

Order — Freedom

Extrovert — Introvert

Thinking — Feeling

Judging — Perceiving

Goals

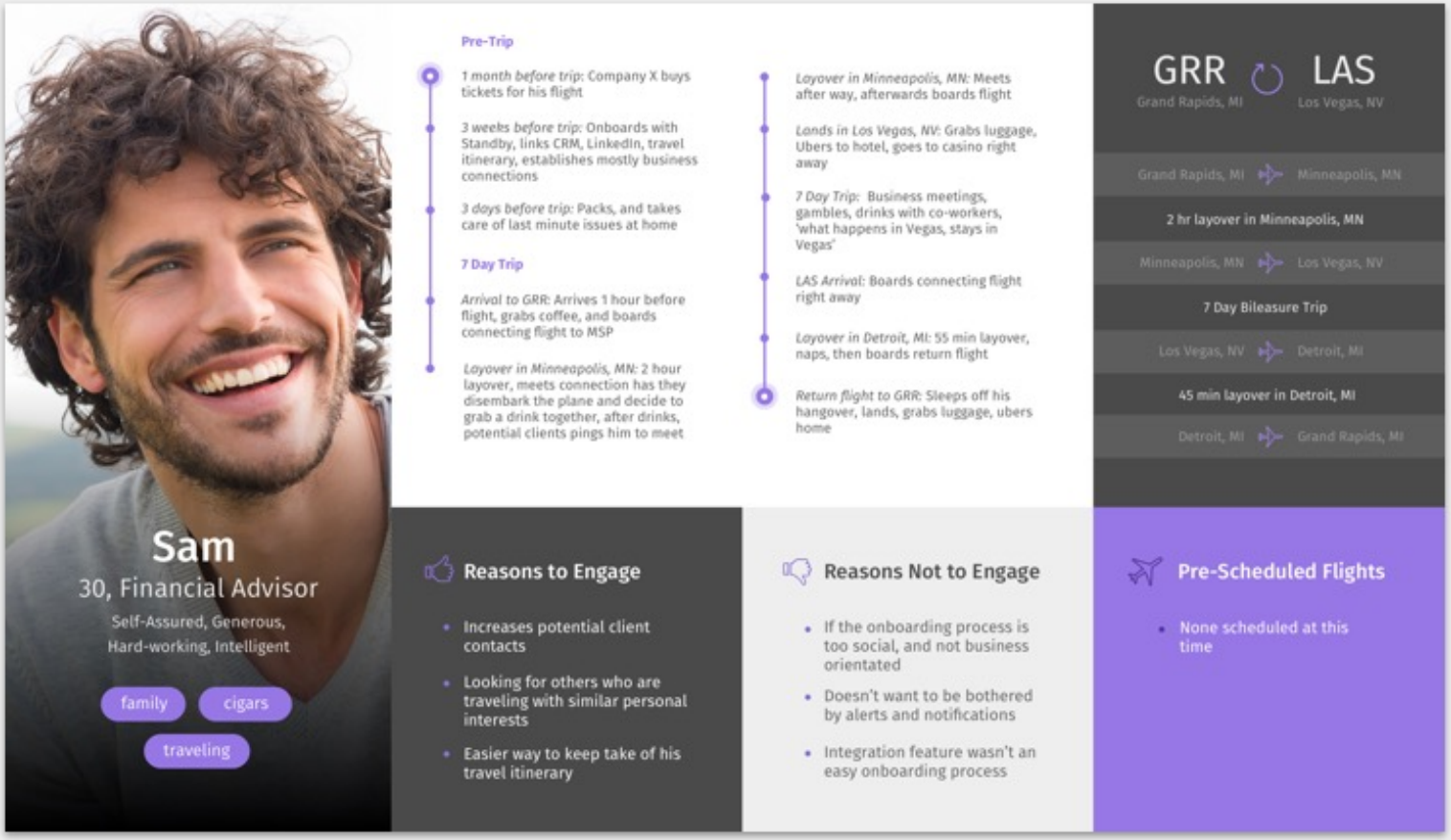
- Easy discovery of news and updates
- News and updates of space science & technology domain
- Comment and share articles
- Save news for later reading

Pain Point

Difficult to find single hub of getting news of news and updates of new things happening around the glob. Need to read on various apps to get news and updates.

"I want easy discovery of news and happing around technology & space science"

Example of Persona



Exercise 1a: check the Personas examples

- **10 minutes**, individually
- Have a careful look to the examples I provided in this lesson as well as in the previous lesson about Personas
- Consider in particular
  - The overall layout
  - The headings of the various sections
  - Any graphical element (e.g., profile picture, logos)

Exercise 1b: design a Persona

- **15 minutes**, individually, pen and paper or any software tool you are familiar with (e.g., powerpoint, draw.io, etc.)
- Think of the personas you are designing for (building on the results of the analysis if you have already conducted it, otherwise grounding on your assumptions)
- Characterize the persona of your primary user (i.e., build his/her profile)
- Design the:
  - **Demographic variables**
  - **Goals**
  - **Motivations**
  - **Needs / desires**
  - **Pain points / frustrations / difficulties**
  - **Preferences**
  - **Skills**
  - **Technology proficiency**
  - **Etc……**

Exercise 1c: compare the designed Personas

- 5 minutes
- In group
- **Compare** your personas with those of your groupmates
- **If they differ, find an agreement**

Exam: not just one persona

- You may have had the time to define the profile of just one persona
- **One persona is not enough, typically a design problem affects more than one category of people**
- Finish the exercise on the personas at home: they are crucial for the exam
- Consider first of all your primary users

Designing the solution

- Now that you have identified the precise **problem** you want to address and the **people** you are designing for…
- …you can start, be creative and propose several ideas of possible **solutions**

1. **Conceptual design**
2. **Concrete design**

- You can use the brainstorming technique

Conceptual and concrete design

- There are two aspects to design:

1. **The conceptual part:** focuses on the idea of a product
   - It involves developing a conceptual model that captures what the product will do and how it will behave
2. **The concrete aspect:** focuses on the details of the design
   - E.g., menu types, haptic feedback, physical widgets, and graphics

Conceptual design

- Is concerned with developing a **conceptual model** of the user-product interaction
- A conceptual model is:
  - An outline of what people can do with a product
  - What concepts are needed to understand and interact with it

Brainstorming

- The classic way of conducting a brainstorming consists of:

1. A first moment of individual **production of ideas**
2. A second moment of **presentation** and discussion of the ideas with the team
3. A third moment of **voting** of the best ideas (ideas can be merged)

Exercise 2: brainstorm solutions

- 15 minutes

1. **Produce ideas** individually:
   - 1 idea per post-it / paper
   - Be visual (i.e., draw), you will explain later what you meant with those lines if that's not clear
   - Produce as many idea as possible
   - Don't censure yourself
2. **Present** your ideas to the group and cluster them for **similarities** (i.e., you will build an affinity diagram)
3. **Discuss** the ideas and **vote** for the best one (complementary ideas can be merged)

Concept

- A concept is a **refined description** of the idea elected to be the solution of your problem
- In the concept, you should describe
  - The problem that it solves
  - How does it solve the problem
  - All the parts that compose it (e.g., a service is made of many touchpoints)
  - How the user can interact with it

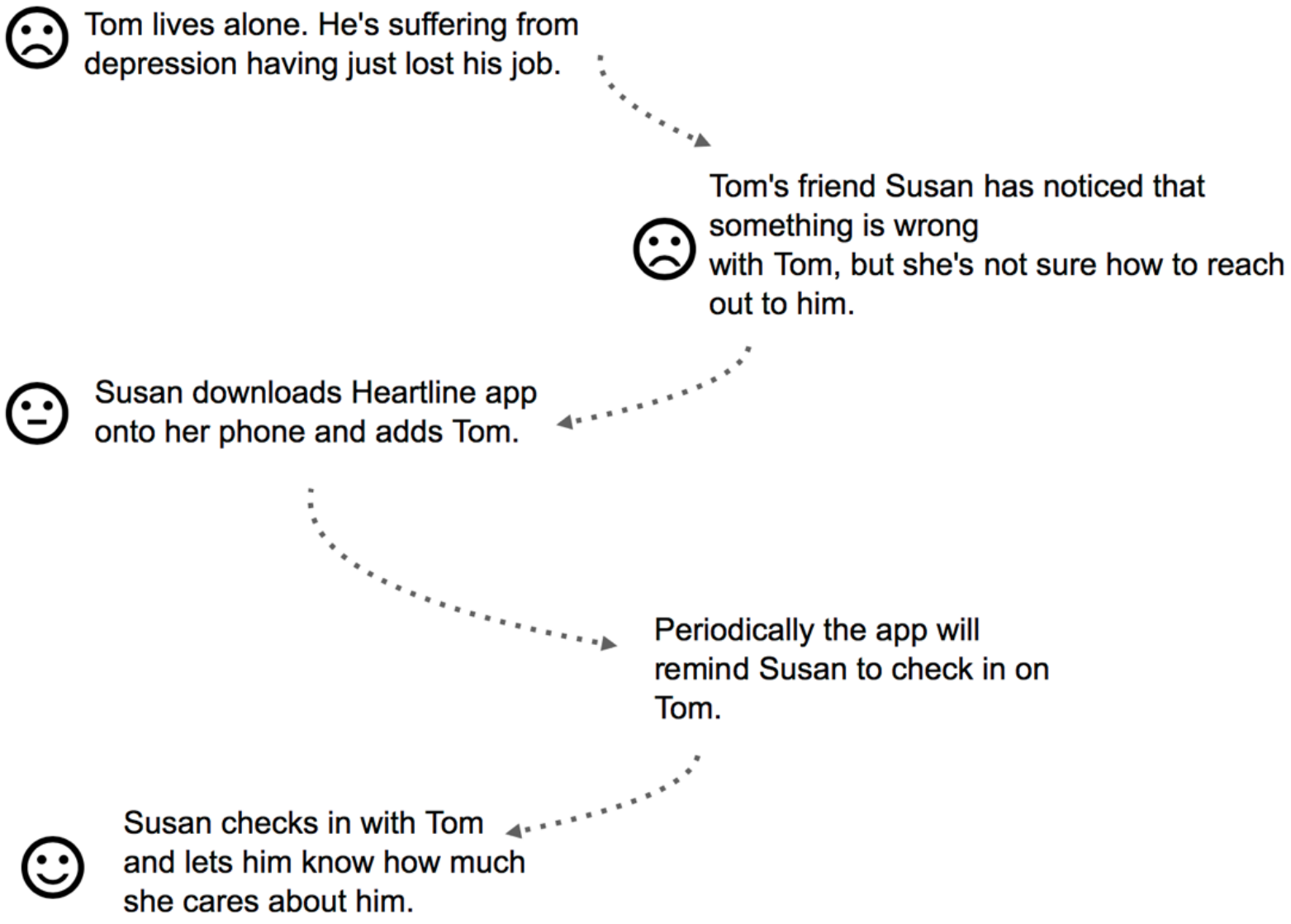
Exercise 3: define the concept of your solution

- In group
- 10 minutes
- Together with your groupmates: write a **paragraph** explaining **precisely** what your solution consists of and how it works
  - Note: you will need to explain this in the exam report
- Please, make sure to check that your solution addresses the problem you have identified

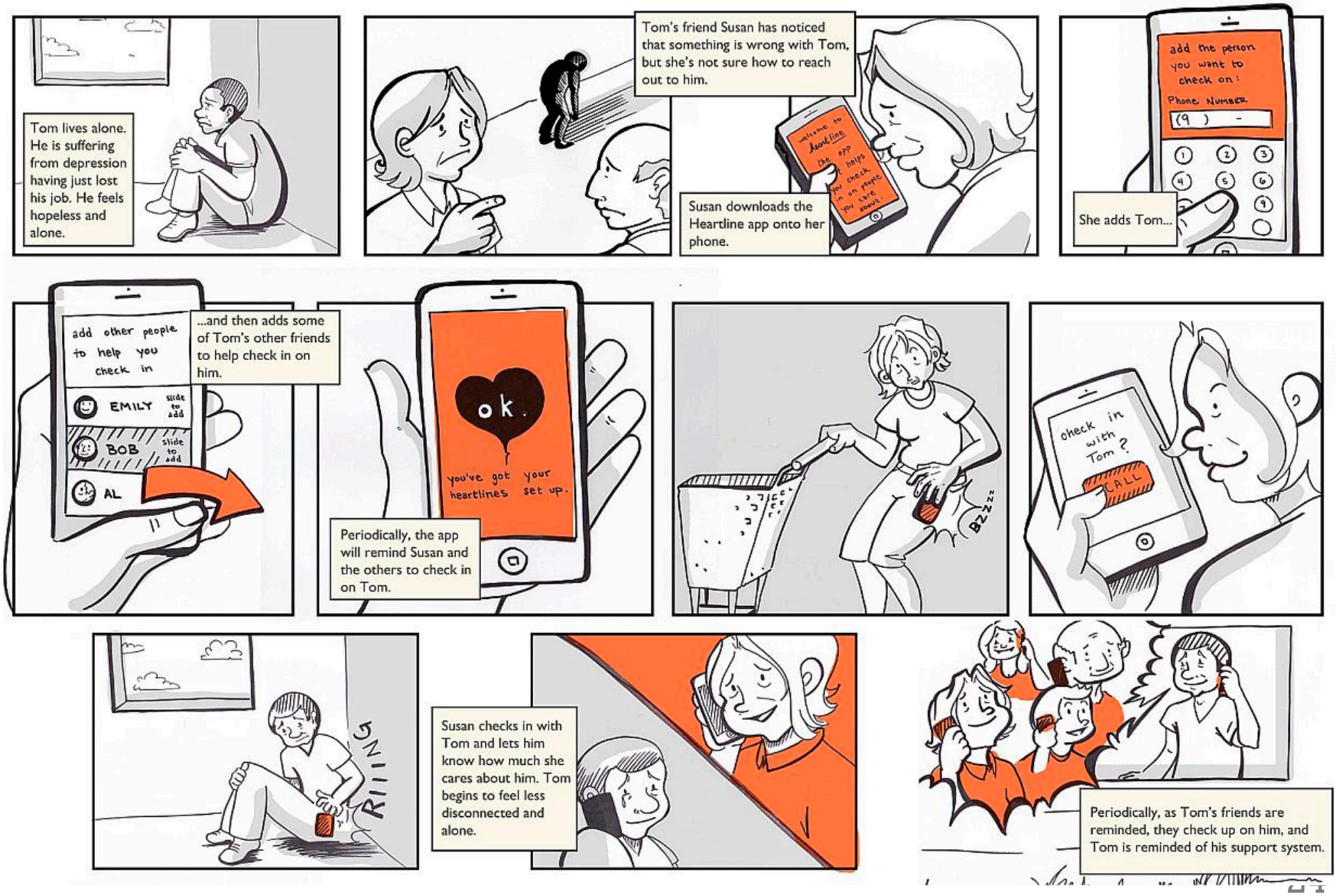
Scenarios

- Scenarios can be either
  - **Activity scenarios**: they show how people's activity is nowadays;
  - **Envisioning scenarios**: projections of how the **future** will be when our technological product will be implemented; specifically, how the lives of our personas would change after the adoption of our product
- Scenarios can be either:
  - **textual** (as a sort of storytelling) or
  - **visual** (e.g., a storyboard, video, etc.).

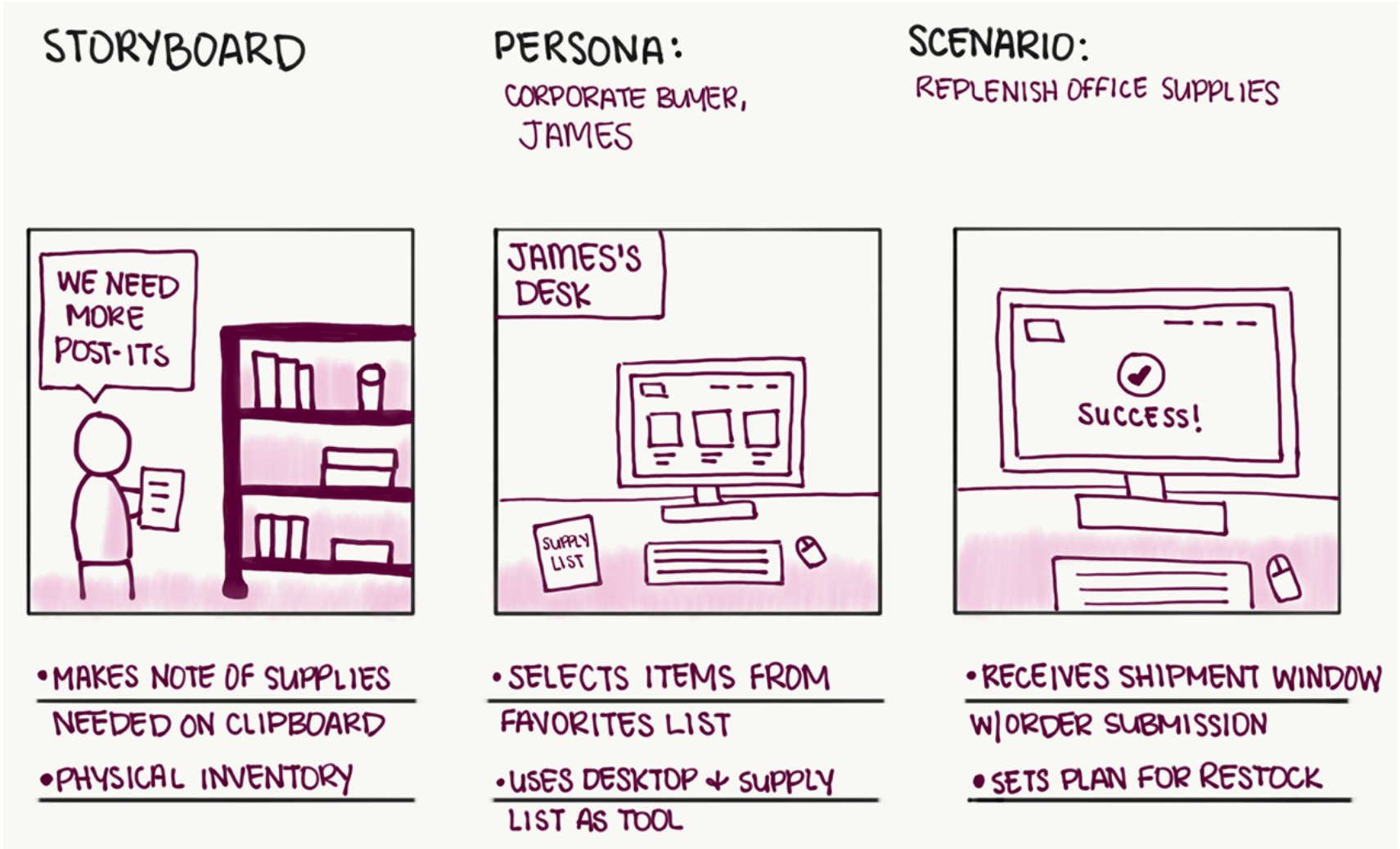
Example of scenario: storyboard



Example of scenario: storyboard



Example of scenario: storyboard



Example of scenario: storyboard



What do you notice?

- Visual scenarios → storyboards (comics and cinema language)
- Dialogues, but most important: captions!
- Use of colours?
- Convey a sense of motion?
- Convey emotions?

What do you notice?

- **Colours** are used to capture attention; they can represent: technology, virtual phenomena (e.g., QR code scanning), direction of movement





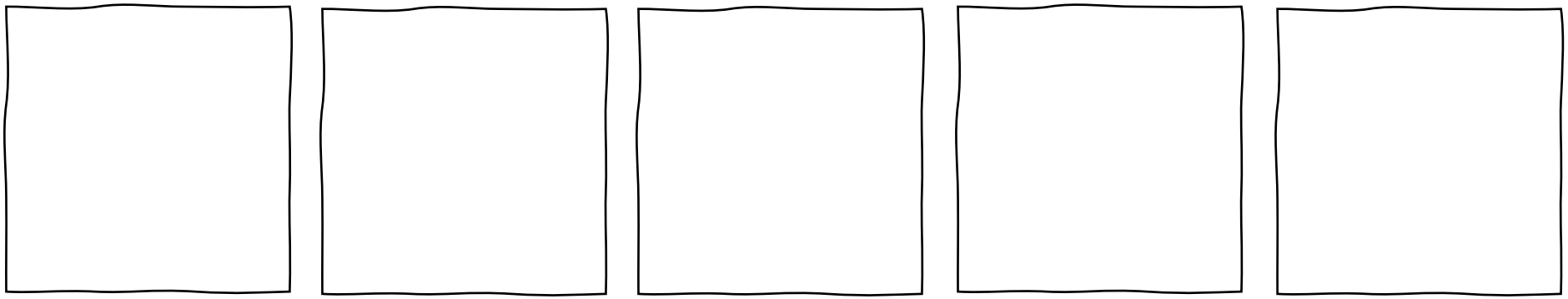
from presentations accompanying the book 'Sketching User Experiences, the Workbook'
by S. Greenberg, S. Carpendale, N. Marquardt and B. Buxton

Please notice

- We are not making art
- **The point is to communicate effectively an idea**
- We should be as essential, minimalistic and simple as possible in our sketches
- **The difficulty stands in being minimalistic and still being able to convey our message**

Exercise 4: draw an envisioning scenario

- 15 minutes, individually
- Draw a scenario of a situation where your persona solves a problem by **using your technological product**
- Use e.g., 5 frames
  - Draw 5 squares like the ones below
  - Put a caption under each frame



Scenarios: video

- Recall that for the exam you can also provide a short video to illustrate an envisioning scenario

Low-fidelity prototypes

- Low-fidelity prototypes may be hand sketches that represent the main functionalities of an interface
- You can also use Axure, but in the next exercise use pencil and paper
  - Note: also for your exam project you can just use Axure, you are free to choose whether using it or create a paper prototype

Exercise 5: sketch a paper prototype

- Individually
- 15 minutes
- Draw 4 screenshots / main stages / functions of your interactive product
- Then compare them with those of your teammates and discuss which one represent your concept better
  - Be careful to decide, *better according to what?* E.g., more intuitive, more beautiful, easy to understand, best position on screen, etc.

Exercise 5: sketch a paper prototype

- If you are planning to design an app, you can use 4 frames of a smartphone to help you respect proportions.



Homework: prototype in Axure

- Start prototyping in Axure the paper prototype you have just created during this laboratory
  - No high level of detail: it is just a low-fidelity digital prototype
  - Start by placing the widgets
  - Then associate the interactions

Bibliography

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